
power
Power
Power
P--W/t
Force
Force = W/t

power

The rate of doing work is called Power. Accordingly, we can talk about instantaneous power and average power.

$$P_{inst.} = \frac{dw}{dt}$$

$$P_{avg} = \frac{\Delta w}{\Delta t}$$

$$P = \frac{\vec{F}.d\vec{s}}{dt}$$

$$P_{inst} = \vec{F}.\vec{v}$$

by differentiating work energy theorem, we get, WET in terms of power i.e. Power done by all the forces is equal to rate of change of kinetic energy.

Que) The position of a particle varies with time as $x = kt^3\hat{i}$.
Find (i) instantaneous power at a general time.
(ii) average power till a general time t. Mass of particle is m.

$P_{inst.} \Rightarrow \quad a = 6kt$

$v = 3kt^2$

$= ma.v$

$= m \times 6kt \times 3kt^2$

$= 18k^2mt^3$

$$P_{avg} = \frac{KE_2 - KE_1}{\Delta t} = \frac{\frac{1}{2}m(v^2-u^2)}{t}$$

$$= \frac{1}{2}\frac{m}{t} \times (9k^2t^4 - 0)$$

$$= \frac{1}{2}\frac{m \times 9k^2t^4}{t}$$

$$= \frac{9}{2}k^2t^3m$$

Or

$$P_{inst.} = \frac{dK_E}{dt}$$

$$= 18mk^2t^3$$

Que) $\vec{r} = A\sin(\omega t)\hat{i} + A\cos(\omega t)\hat{j}$, mass is m
Find instantaneous and avg. power.

Using WET $\quad v = A\cos(\omega t)\hat{i} + A\sin(\omega t)\hat{j}$

$(\vec{v}.\vec{v}) \quad v^2 = A^2\cos^2(\omega t)\hat{i} + A^2\sin(\omega t)\hat{j}$

$v^2 = A^2$

$u^2 = A^2$

$\Delta KE = \Delta W = \frac{1}{2}m(v^2 - u^2) = 0$

$P_{avg} = 0$

$P_{inst} = 0$

123